INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

Salt Lake Campus

Under University of Engineering & Management, Kolkata

Degree: BCA , Stream: BCA

Term – I, First Semester Examination, September -2023

Subject Code: BCAMDBC101

Subject Name: Digital Electronics

Full Marks:30 Duration:1 Hour

Part - A

Attempt 5 questions

Each question carries 2 Marks (2 X 5)

1. Explain how a NAND gate can be used to design an OR gate. 2

or

1. Describe under what conditions will the output of an NOR gate be 0? 2

2. Examine the difference between latch and flip-flop. 2

or

2. Determine $(-46)_{10}$ into 8 bit binary number using signed binary number representation. 2

3. Calculate $(225.225)_{10}$ into binary number. 2
and $(623.77)_8=(?)_{16}$

or

3. Illustrate the 9's complement and 10's complement of $(10000)_{10}$ and $(00000)_{10}$. 2

4. Describe what is meant by duality in Boolean algebra? 2

or

4. Identify the EX-NOR gate? Write its truth table. 2

5. Modify the following: 2
   - $(2648)_8=(?)_2$
   - $(2AC5.D)_{16}=(?)_8$

or

5. Demonstrate De-Morgan's Law with expression. 2

Part - B

Attempt 2 questions

Each question carries 5 Marks (5 X 2)

6. Assess and prove the following Boolean expression 5

$(A + B)(\bar{A}\,\bar{C} + C)\overline{(\bar{B} + AC)} = \bar{A}\,B$

or

6. Evaluate the truth table of S-R flip-flop using NAND gate. 5

7. Predict the result of addition and multiplication operation of the following numbers in the given base without converting to decimal: $(367)_8$ and $(715)_8$ 5

or

7. Calculate the following: 5

- $(110111)_2 - (11011)_2$ [Using 1's compliment]
- $(+49)_{10} + (-37)_{10}$ [Using 2's compliment]
- $(10110.1)_2 - (1100.01)_2$ [Using binary subtraction]
- $(20)_{10} - (1000)_{10}$ [using 10's compliment]
- Divide $(111111)_2$ by $(1001)_2$

Part - C

Attempt 1 question

Each question carries 10 Marks (10 X 1)

8. Illustrate the working principal of full adder circuit and derive its expression with the help of K-Map simplification. 10

or

8. Analyse the logical expression $ABCD + \bar{A}BC + \bar{B}\bar{C}$ on a 4-variable K-map; obtain the simplified expression from the map and draw the circuit diagram. 10